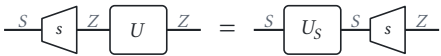
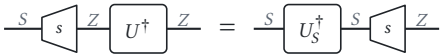
(a) U maps S into S: $Us = sU_S$



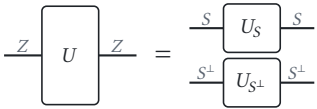
(b) step 1: $U_S^\dagger U_S = s^\dagger U^\dagger U s = s^\dagger s = I_S$

an isometry of S into itself, hence onto, since $\dim S < \infty$

(c) step 2: $U^\dagger s = sU_S^\dagger$, so $U^\dagger$ maps S into S



(d) step 3 and the conclusion: $U = U_S \oplus U_{S^\perp}$



$Z = S \oplus S^\perp$, and the two wires never meet